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Fetal cerebral biometry: normal parenchymal findings and ventricular size

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Abstract Assessing fetal cerebral biometry is one means of ascertaining that the development of the fetal central nervous system is normal. Norms have been established on large cohorts of fetuses by sonographic and neurofetopathological studies. Biometric standards have been established in MR in much smaller cohorts. The purpose of this paper is to analyse methods of measuring a few parameters in MR [biparietal diameter (BPD), fronto-occipital diameter

(FOD), length of the corpus callosum (LCC), atrial diameter, transverse cerebellar diameter, height, antero-posterior diameter and surface of the vermis] and to compare US and MR in the assessment of fetal cerebral biometry.

Keywords Fetus · Brain · Magnetic resonance · Biometry · Brain growth and development

Introduction

There are few means of ascertaining that the development of the fetal central nervous system is normal. Assessing fetal cerebral biometry is one of these means. Biometrical norms have been established in neurofetopathology but these studies are based on the analysis of fixed brains. Many cerebral biometric data have been provided by sonographic studies and have been established on large cohorts of fetuses. For a few years, MR biometric data have also been available. However, unlike US, MR is not a routine examination in antenatal diagnosis and for this reason, standards have been established in much smaller cohorts than for US (225 fetuses [1]). The purpose of this review is to evaluate the contribution of US and MR in the assessment of fetal cerebral biometry.

In current practice, several measurements are commonly obtained:

- Head size may be measured: biparietal diameter (BDP) and head circumference (HC) in US and fronto-occipital diameter, cerebral and bone BPD in MR.

- Some cerebral structures may also be measured: corpus callosum (length, thickness), ventricles (atrial diameter), cerebellum (transverse cerebellar diameter, height, anteroposterior diameter and surface of the vermis).

Technique

In US, measurement of some cerebral structures may be difficult to obtain in some conditions related mostly to patient obesity and position of the fetal head.

Measurement of cerebral structures with MR is independent of these parameters. It is obtained with the acquisition of T2-weighted sequences. MR has a better contrast resolution than US but a worse spatial resolution.

Head size (Table 1)

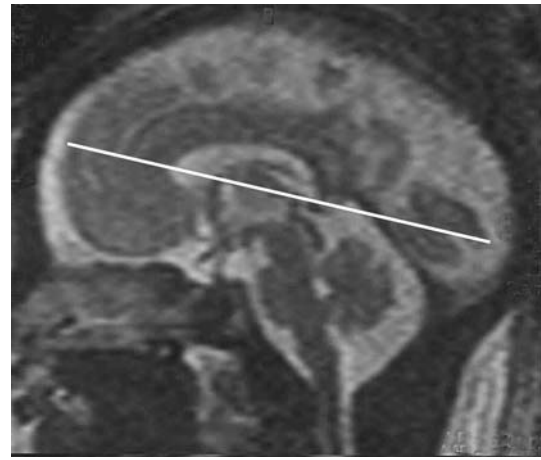
In US, BPD and HC are measured on a transthalamic slice: on this slice, the cavum septi pellucidi, the third ventricle, the thalami and the sylvian fissures must be visualised. Both

Table 1 Main MR biometric data for the fronto-occipital and biparietal diameters, the corpus callosum and the cerebellum

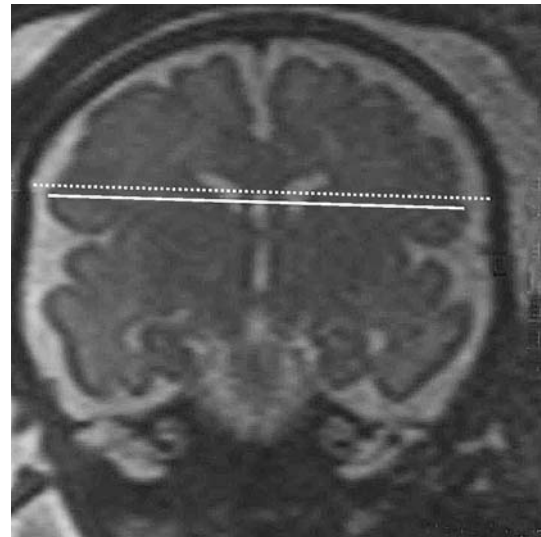
		50th Percentile (mm)
Frontal-occipital diameter	27 WG	80
	31 WG	90
	35–36 WG	100
	WG	
Cerebral biparietal diameter	22–23 WG	50
	27 WG	60
	31 WG	70
	34–35 WG	80
	WG	
Length of the corpus callosum	22–23 WG	30
	29 WG	35
	33 WG	40
	WG	
Transverse diameter of the cerebellum	22–23 WG	25
	27 WG	30
	29 WG	35
	32 WG	40
	35–36 WG	46
	WG	
Height of the vermis	22–23 WG	11
	27 WG	16
	32 WG	18
	34 WG	20
	WG	
Anteroposterior diameter of the vermis	22–23 WG	8
	27 WG	10
	31 WG	12
	33 WG	14

From [24]

BPD and HC are bone measurements. Actually, pericerebral space is visible in US only when it is markedly enlarged so that in current practice, it is impossible to measure cerebral BPD in a reproducible way. Measurement of HC takes account of head shape. In current practice, the fronto-occipital diameter (FOD) is not measured in US examinations but standards have been published [2]. In MR, it is possible to evaluate bone and brain measurements. The fronto-occipital diameter is the distance between the extreme points of the frontal and occipital lobes on a midline sagittal slice (Fig. 1). Bone and cerebral BPD are measured on a coronal slice at the level of the temporal horns of the lateral ventricles (Fig. 2). The difference between both measurements corresponds to the pericerebral space [1]. It is interesting to take account of this value in macrocephalies. MR makes it possible to differentiate between macrocrania which results

**Fig. 1** Fronto-occipital diameter is measured on the midline sagittal slice, as the distance between the extreme points of the frontal and occipital lobes

from an excessive growth of the skull (bone BPD is increased with an enlargement of the pericerebral space and a normal cerebral BPD) and macrocephaly or megalencephaly which is characterised by an excessive brain volume (bone BPD and cerebral BPD are increased whereas pericerebral space is thin). Sonographic values are somewhat higher than MR ones when MR measurements are obtained in that manner because the thickness of the cranial vault is not measured with MR. Conversely, if measurements of BPD and HC are obtained in a similar manner in US and MR, a significant correlation is found between both measurements [3].

**Fig. 2** Bone and cerebral biparietal diameters (BPDs). They are both measured on the coronal slice at the level of the temporal horns of the lateral ventricles. Bone BPD (*dotted line*) is the distance between the two internal tables of the skull. Cerebral BPD (*plain line*) is the greatest transversal diameter of the brain. The difference between both measurements is the size of pericerebral space

Corpus callosum (Table 1)

The length of the corpus callosum (LCC) can be measured from the genu to the posterior aspect of the splenium (Fig. 3). In US, the corpus callosum is hypoechoic and standard values have been established on transfontanellar slices [4].

In MR, the corpus callosum is T2 hypointense. US and MR values are concordant [5]. Growth of the corpus callosum is linear during the second trimester. The LCC/FOD index decreases throughout pregnancy (44.6% at 22 weeks, 41% at 38 weeks), which indicates that the corpus callosum's growth is proportionally slower than the brain's. The spatial resolution of MRI is worse than in US so that it is impossible to evaluate precisely the thickness of the corpus callosum.

Lateral ventricles

In US, the atrial diameter is measured on a transventricular view, which is obtained superior and parallel to the trans-thalamic view, at the level of the ventricles [6]. On this view, the atrial diameter is measured in its largest axial dimension in the cerebral hemisphere farther from the transducer [7].

This technique provides a reliable and reproducible measurement. During the last two trimesters of pregnancy, the atrial diameter is relatively stable [8, 9] with a mean value of 7.6 ± 0.6 mm [standard deviation (SD)] giving a measurement of 10 mm as +4 SD above the mean [10]. A difference may exist on the size of the atrial diameter between the sexes. This value would be slightly higher for males [11]. For other authors, this difference is small enough not to be clinically relevant [12]. The problem with this technique is that the proximal ventricle is almost always obscured by nearfield artefact so that the ventricle closest to the probe is not properly analysed. It has been suggested [13] to position more caudally the probe on the fetal head so that the plane is

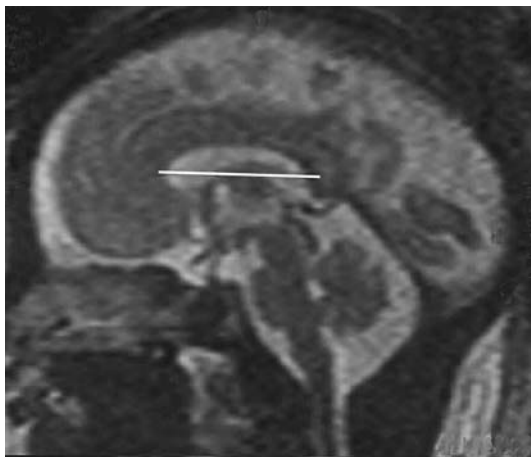


Fig. 3 Length of the corpus callosum. It is measured on the midline sagittal slice, from the genu to the posterior extremity of the splenium

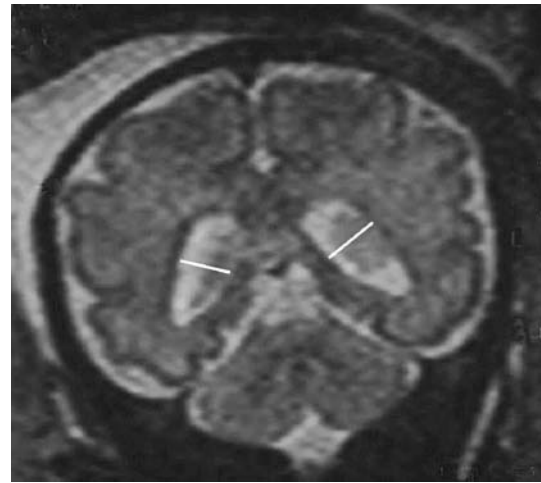


Fig. 4 Lateral ventricles. The atrial diameter is measured on the coronal slice at the level of the atria, on an axis perpendicular to that of the ventricle, at mid-height of the atrium

angled cephalad. This oblique view does not provide a symmetric analysis of both lateral ventricles. It seems better to evaluate the atrial diameter on a coronal view and if possible, on a transfontanellar view. The discrepancy between axial and coronal measurements does not exceed 2 mm [14] and the closest ventricle is not hindered by nearfield artefact. Whatever the technique, the cursors must be placed on the medial and lateral walls of the atrium.

In MR, we suggest to measure the atrial diameter on a coronal slice at the level of the atria, on an axis perpendicular to that of the ventricle, at mid height of the ventricle [1] (Fig. 4). MR has the advantage over US that both ventricles (deepest and most superficial) are equally clearly visible and measurement is possible whatever the position of the fetal head. Moreover, due to a better contrast resolution, the walls of the ventricle are more clearly delineated with MR than with US so that the measurement is more precise and reliable. In a series of 23 fetuses without CNS anomalies, atrial diameter seems to be constant in MR throughout pregnancy (from 14 to 40 weeks). In this study, MR measurements were slightly smaller than mean US measurements and the ventricular atria cutoff value defined as 2 SDs above the mean is >10 mm [15].

In this study, atrial image measurement was obtained in the axial plane. In our experience, we evaluate the atrial diameter on a coronal slice (in both US and MR) and we do not find a significant discrepancy between US and MR in the measurement of both lateral ventricles. In US, the coronal slice is obviously more reliable than the axial slice for measuring the most superficial ventricle.

Third and fourth ventricles

In US, the average diameter of the third ventricle is relatively constant (1 mm) before 28 weeks and is never higher than 1

mm after that stage [16]. It is considered to be abnormal when the diameter exceeds 3.5 mm.

Average values are a little higher in MR but do not exceed 4 mm [17].

This small difference between US and MR is probably explained by the better contrast resolution of MR which allows a sharper delineation of the third ventricle than with US. However, it must be kept in mind that the small size of this structure is responsible for the great uncertainty of its measurement whatever the technique used.

On axial slices, the anteroposterior diameter of the fourth ventricle is 3.5 ± 1.3 mm in US [18]. In MR, the anteroposterior diameter is measured on a midline sagittal slice and its average value is 4.5 ± 2.5 mm.

The cerebellum (Table 1)

The *transverse cerebellar diameter* is held to be a good marker for gestational age [19]. The hemispheres are hypoechoic with echogenic margins. In US, the transverse cerebellar diameter is usually measured on an axial slice but it can also be measured on a coronal slice if the axial slice is not available because of the fetal position. There is no significant difference between both measurements [20].

In MR, the transverse cerebellum diameter is measured on a posterior coronal slice at the level of the atria (Fig. 5). This measurement is superimposable to that obtained by US except at the end of pregnancy. In late stages, the artefacts caused by the petrous part of the temporal bone are responsible for a poor visibility of the margins of the cerebellum and hence a poor precision of sonographic measurement which explains the discordance between US and MR measurements.

MR measurement remains reliable throughout pregnancy because there are no artefacts related to bone structures.

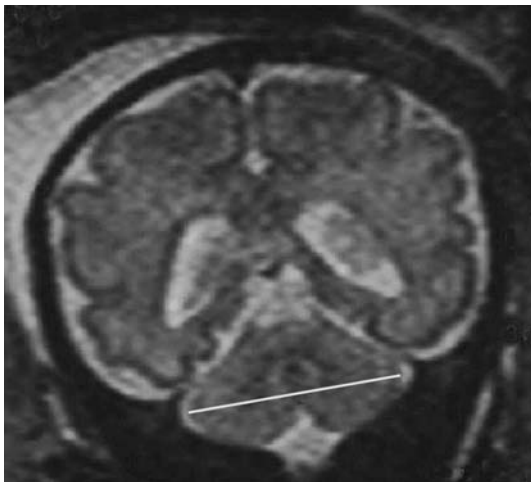


Fig. 5 Transverse cerebellar diameter. It is measured on the posterior coronal slice at the level of the atria

The vermis In US, the vermis is characteristically hyper-echoic compared to the hypoechoic cerebellar hemispheres. Its anteroposterior diameter, its height and its surface can be measured on a midline sagittal slice and norms have been established in premature and term babies [5, 21–23]. In Birnholz's study, the surface of the vermis was measured on transfontanellar slices and the vermis was assimilated to an ellipsis. In our study (Fig. 6), measurement of the surface of the vermis was performed with MR and was made by manual computer planimetry on the midline sagittal slice. Hence, the technique and conditions of measurement are very different with US and it seems difficult to compare the

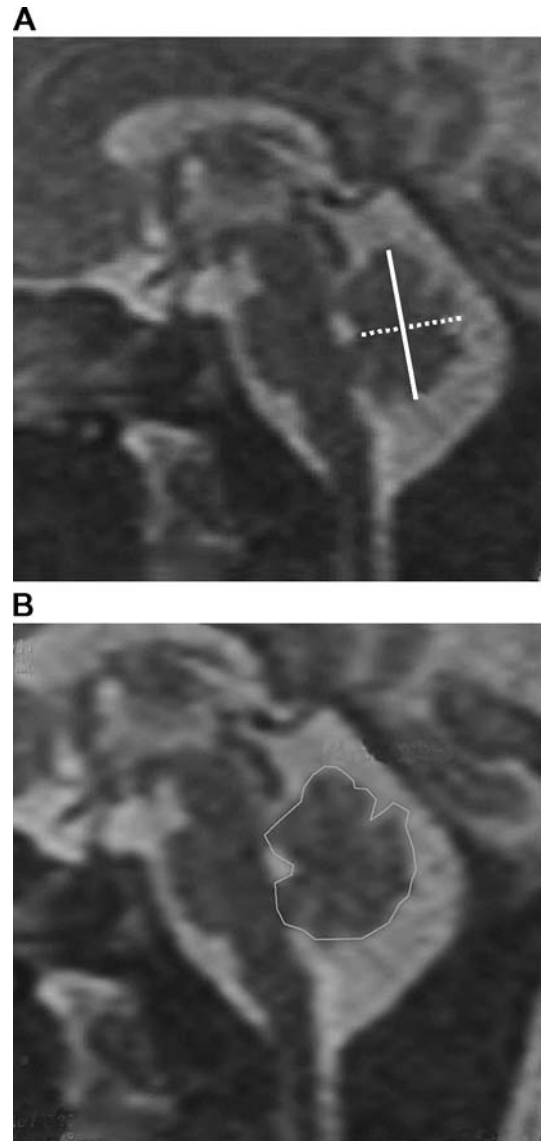


Fig. 6 Vermis. **A** Height (plain line) and anteroposterior diameter (dotted line) of the vermis are measured on the midline sagittal slice. **B** Measurement of the surface of the vermis is made by manual computer planimetry on the midline sagittal slice

results. The spatial resolution of MR currently does not allow each of the nine lobules composing the vermis to be considered separately.

Conclusion

MRI contributes to a better understanding and analysis of fetal brain development. Assessing fetal biometry is one

means of ascertaining normal brain development. Standards of reference can be used to assess normality of biometry. However, it must be kept in mind that, as after birth, individual variations in this chronology have to be taken into account in any analysis of the fetal brain.

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